

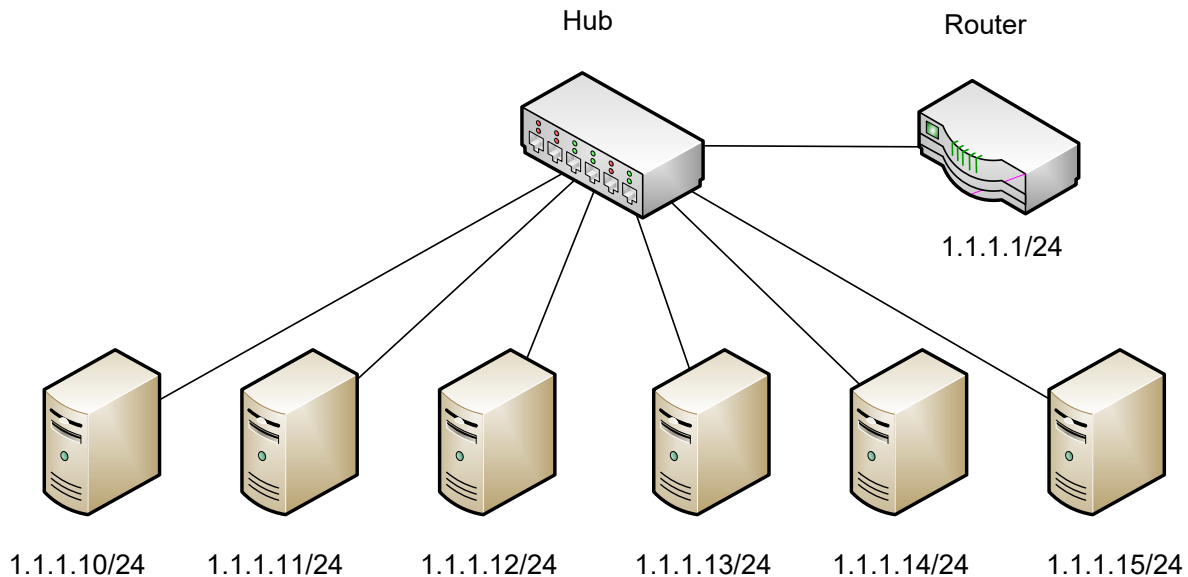
Lab 4

Network Traffic Sniffer

Objective:

The purpose of this lab is for the student to observe network traffic. Students will utilize a network sniffing software to capture and view network packets.

Diagram:



Components:

6 – PC's with an Ethernet NIC

1 – Hub

1 - Router

Procedure:

1. Connect all PC's together and put their static IP addresses on them.
Verify connectivity by pinging each machine.
2. Run the Wireshark program (as Administrator). Do this by right-clicking the Wireshark icon and then clicking "Run as Administrator".
3. Select Capture and then Options
4. Make sure you have selected the proper interface. It will display the IP address of the interface you are using.
5. Select start and you should observe some traffic being captured.
6. Select capture and then stop to end the capture
7. The program is broken down into 3 windows. Describe what you see in each window:
 - a. Top
Packet list pane
 - b. Middle:
Packet details pane
 - c. Bottom:
Packet bytes pane
8. Select Capture and then Options

9. Under the capture filter type in **host 1.1.1.10**
10. Select start and then bring up a command prompt and ping 1.1.1.10
11. Once the ping has completed stop the capture and view the packets
12. How many packets do you have?
18 (a few extra at the tail end; 8 for the ping process)
13. What type of packets are the first two?
ARP & ARP Response are 9 and 10; 1-8 are ICMP ping requests and replies
14. Does a ping send 4 packets and then wait for a reply or is it a request then reply?
Request & reply; 8 packets total
15. What data is actually sent in an icmp packet?
It sends the alphabet, or most of it
16. Start a new capture with a filter of host 1.1.1.1
17. Once the capture is completed view the traffic.
18. After the arp request what are the three packets that begin the tcp session?
SYN, SYN/ACK, ACK

19. Find the telnet password and the enable password based on what the user typed in

Telnet Password: TelnetPassword

Enable Password: EnablePassword

20. Find the telnet password and the enable password based on the user receiving the configuration of the device.

21. The instructor will change both passwords on the device.

22. Start a new capture with a filter of host 1.1.1.1. Once the capture is completed view the traffic.

23. Find the telnet password and the enable password

Telnet Password: NewTelnetPassword

Enable Password: NewEnablePassword

24. What did you learn by completing this lab?

I learned how to manually set the IP address to a Windows computer

I learned how to use WireShark to capture network traffic

I learned how to use the Analyze function to reconstruct packets sent via Telnet protocol and analyze them

25. Draw the lab diagram below (PC's and hub connections) Label the IP addresses, and subnet masks of each device. Feel free to use packet tracer or Visio, but do not copy and paste the diagram at the beginning of the lab. Screen shot your diagram and paste it below:

